Multi-Agent Financial Advisor

Mid-Term Progress Report

Fordham University · MSQF Capstone 2026
Mentor: Ocean Li

Aidan Altorelli James Huang Niharika Madana

July 13, 2026

1 Research Question & Scope

Can a multi-agent AI system produce personalized, compliance-cleared investment portfolios that account for a client's *total* wealth, financial capital and human capital combined, in a way that is quantitatively rigorous, academically grounded, and regulatorily defensible?

The project operationalizes the Human Capital framework of Ibbotson, Milevsky, Chen & Zhu (2007), which treats a client's future earning stream as a financial asset with its own risk profile. A tenured biology professor's salary behaves like a bond, stable, low-beta, predictable while a software developer with RSUs has income that co-moves with the equity market. These career-embedded exposures must be subtracted from the client's equity budget before a financial portfolio is constructed. The system formalizes this as:

$$portfolio\_equity\_target = \underbrace{\frac{FC + HC\,(1-\sigma)}{W}}_{\text{effective risk budget}} - \underbrace{\frac{HC}{W}}_{\text{implicit equity exposure}} \cdot \beta$$

where FC is financial capital, HC is the present value of future earnings (discounted at the live FRED DGS10 rate), σ is earnings volatility, W is total wealth, and β is the income-to-market beta.

The pipeline covers nine BLS occupational personas spanning bond-like, mixed, and equity-like human capital profiles, generating a compliance-cleared `AdvisorPackage` for each.

2 System Architecture & Shared Infrastructure

Five specialized agents are coordinated by an Orchestrator and communicate through Pydantic v2 contracts defined in `contracts.py` . LLMs participate only in rationale text and reasoning traces; all quantitative outputs are deterministic.

Agent	Entry Point	Output Contract
Profile	`run_profile_agent()`	`ProfileAgentOutput`
Research	`run_research_agent()`	`MacroRegimeSnapshot`
Allocation	`run_allocation_agent()`	`AllocationAgentOutput`
Risk	`run_risk_agent()`	`RiskAgentOutput`
Compliance	`run_compliance()`	`ComplianceAgentOutput`
Orchestrator	`run_pipeline()`	`AdvisorPackage`

3 Agent Implementations

3.1 Profile Agent

The Profile Agent is the first quantitative stage: it converts a raw occupational persona into a validated `ProfileAgentOutput` that quantifies how much of a client's *total* wealth is already implicitly exposed to equity risk through their career. It is deterministic and single-pass, with no LLM calls and no iterative refinement; every field is derived by formula and checked at construction.

Data and valuation. Four sources feed the agent, each cached to parquet: FRED `DGS10` (the live 10Y Treasury yield used as the human-capital discount rate, 4.4% fallback), BLS OES (May 2023 salary percentiles by SOC code), BLS ECEC (Q1 2026 occupation-group bonus rates), and the Federal Reserve SCF (2022 median financial assets by age and income quartile). Effective annual earnings, base salary inflated by the occupation's bonus rate, are discounted as an annuity over the years n to age 65, and income-equity β/ρ come from a calibrated three-tier lookup (bond-like 0.05/0.10, mixed 0.35/0.40, equity-like 0.90/0.75; Ibbotson et al. 2007; Davis & Willen 2000):

$$HC = \left[\text{Salary} \times (1 + \text{bonus})\right] \cdot \frac{1 - (1 + r)^{-n}}{r}, \qquad \text{PET} = \tfrac{FC + HC(1-\sigma)}{W} - \tfrac{HC}{W}\,\beta.$$

The calibrated lookup is used rather than an OLS income-on-market regression, which BLS cross-sectional wages cannot support without fabricating income shocks.

Categorization and validation. The income-stability tier that sets σ and the human-capital type is encoded as a reproducible scoring rule recovering eight of the nine occupation tiers from the signals present in the data, with the ninth (Sales Manager) a documented override; an import-time integrity check binds every label to its justification (`INCOME_STABILITY_BASIS`). Nine occupations (three per tier, 27 in extended mode) each pass four Pydantic validators, holdings sum to 1.0, total-wealth consistency, the implicit-exposure identity, and β/type consistency, before leaving the agent.

PROFILE AGENT – Nine-Persona Output (BLS p50, discount rate r = 4.4%) [source: agents/profile/profiles_all.json]

Occupation	Career	Age	FinCap	HumanCap	TotWealth	HCtype	β	IEE	RiskBud	EqTarget
Biology Professor	Academia	47	200,000	1,095,155	1,295,155	bond-like	0.05	4.2%	95.8%	+91.6%
Registered Nurse	Healthcare	38	90,000	1,440,907	1,530,907	bond-like	0.05	4.7%	95.3%	+90.6%
Compliance Officer	Government	42	90,000	1,170,712	1,260,712	bond-like	0.05	4.6%	95.4%	+90.8%
Lawyer	Legal	45	200,000	2,066,707	2,266,707	mixed	0.35	31.9%	81.8%	+49.9%
Mechanical Engineer	Engineering	41	90,000	1,579,025	1,669,025	mixed	0.35	33.1%	81.1%	+48.0%
Financial Analyst	Finance	40	90,000	1,646,737	1,736,737	mixed	0.35	33.2%	81.0%	+47.8%
Software Developer	Technology	38	90,000	2,296,908	2,386,908	equity-like	0.90	86.6%	61.5%	−25.1%
IT Manager	Technology	44	90,000	2,537,970	2,627,970	equity-like	0.90	86.9%	61.4%	−25.5%
Sales Manager	Sales	44	90,000	1,958,389	2,048,389	equity-like	0.90	86.0%	61.8%	−24.2%

IEE = Implicit Equity Exposure = HC_share × β | Portfolio Equity Target = Effective Risk Budget − IEE
Bond-like → EqTarget > +90% | Mixed → ~+48–50% | Equity-like → NEGATIVE (career already over-exposed to equity)

Figure 1: Profile Agent output for all nine BLS-derived personas (p50 salary, $r = 4.4\%$): wealth breakdown, implicit equity exposure, effective risk budget, and portfolio equity target by human-capital tier.

Future Directions

- Formal documentation of the income-volatility σ tier values (currently calibrated), paralleling the β/ρ citations.
- Dynamic β estimation from panel income data (e.g. PSID) to replace the static tier lookup; defined-benefit pension valuation for bond-like personas.
- Expanding beyond nine SOC codes and incorporating the SCF 2025 update.

3.2 Research Agent

The Research Agent runs independently of client data, on macroeconomic series alone, and produces a single validated `MacroRegimeSnapshot` that anchors the Allocation and Risk agents to a historically grounded macro state rather than to raw signal noise. It is deterministic with no LLM calls.

Features and segmentation. Thirteen FRED series (1995–2025, monthly) are transformed to stationary signals, z-scores for level series (yield curve, spreads, CPI, breakevens, VIX, industrial production, GDP), rate-of-change for fed funds and unemployment, and first differences for the 5Y/30Y yields—leaving 274 observations after breakeven history begins (2003-02 onward). PELT (`ruptures`, RBF kernel, penalty 10) then identifies structural breaks with no date assumptions, producing five raw macro segments (breaks at 2004-09, 2008-01, 2014-09, 2020-12).

Classification and output. A K-means pass over segment fingerprints is a diagnostic check only; the load-bearing labeller is an XGBoost classifier (100 estimators) trained on five hand-selected anchor windows, mapping every month to one of five business-cycle regimes (Hamilton 1989; Clarida, Galí & Gertler 1999), followed by a six-month majority-vote smoother. Every sequence row is validated against `RegimeRecord` before disk. The most recent snapshot (2025-12) reads Late-Cycle Expansion at confidence 0.902 with no regime change; CRSP value-weighted returns fall in monotonic economic order from Early Recovery (+1.54%/mo) to Inflation Shock (−0.29%/mo), volatility peaking in the crisis regime.

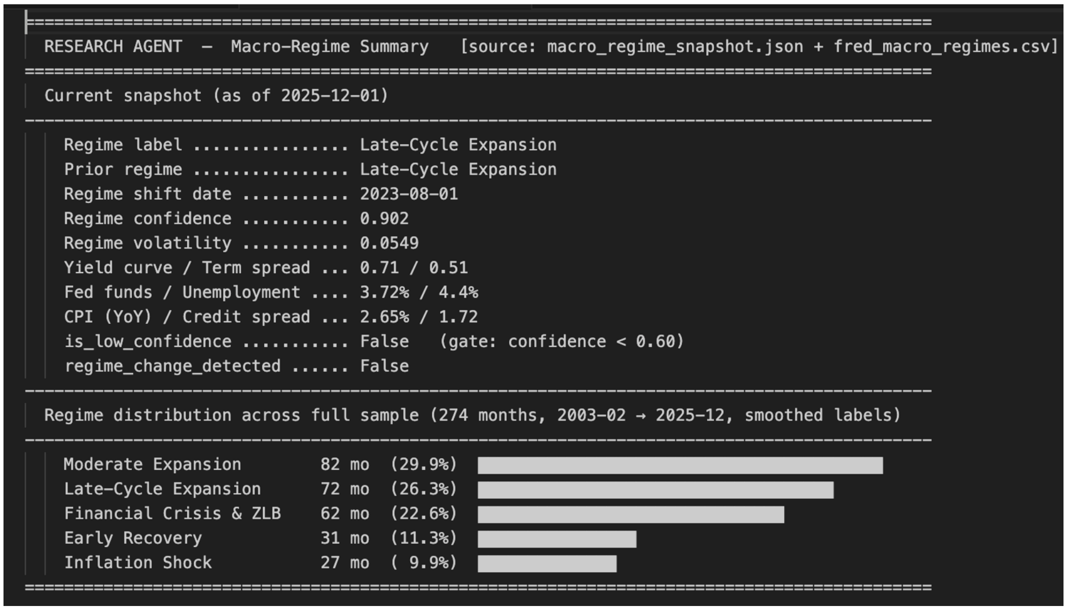

Figure 2: Current macro-regime snapshot (2025-12: Late-Cycle Expansion, confidence 0.902) and the distribution of the five academic regimes across the 274-month sample.

Future Directions

- Formal HMM/GMM benchmark (`model_comparison.py`); preliminary runs favor XGBoost as the only model recovering all five regimes.
- A regime transition-probability matrix, confidence calibration, and an out-of-sample regime-validation harness.

3.3 Allocation Agent

The Allocation Agent (`run_allocation_agent()`, deterministic core in `agents/shared/core/allocation.py`) converts a `ProfileAgentOutput` into a portfolio recommendation over a 24-ticker ETF universe (4 broad equity, 7 fixed income, 2 real assets, 10 GICS sector SPDRs, 1 cash proxy). Only the rationale is LLM-authored (Claude, Touchpoint 1); every weight and statistic is computed before the model sees a number.

Optimization pipeline. An annualized covariance matrix is built from CRSP monthly excess returns aligned to Fama-French factor months, feeding CAPM equilibrium returns $\pi = \delta \Sigma w_{mkt}$ from market-cap weights. Each ticker's excess return is regressed on the four Fama-French factors to build the view matrix P (identity), view returns Q, and view uncertainty $\Omega = \mathrm{diag}(\sigma^2_{resid})/\tau$. The Black-Litterman posterior (μ_{BL}, Σ_{BL}) then feeds two SLSQP-constrained solves, an equilibrium-only baseline w_{eq} and a posterior-tilted w_{BL}, enforcing single-name (10%), sector (20%), economic-sector (25%), and employer concentration limits.

Human-capital overlay. Financial wealth is split into risky and safe sleeves with the BMS (1992) risk-free-HC formula and its Campbell-Viceira (2002) risky-HC extension,

$$w_{fin} = \alpha \left(1 + \frac{H}{W}\right) - \frac{H}{W}\,\beta, \qquad \alpha = \frac{\mu - r_f}{\gamma \sigma^2},$$

where α is the Merton (1971) optimal risky share and γ the risk-tolerance-implied risk aversion coefficient, then *capped at portfolio_equity_target* (§1) so career-plus-portfolio equity exposure never exceeds the client's risk budget. Each ticker's weight is decomposed into equilibrium baseline, view tilt, and human-capital offset; portfolio statistics blend the risky sleeve with a zero-volatility safe sleeve so expected return, volatility, Sharpe, and factor exposures describe the whole portfolio rather than the risky sleeve alone.

Integration testing this period found two contract-conformance gaps, both now fixed: `human_capital_type` was computed by the Profile Agent but never carried by the Allocation Agent's internal schema (`HumanCapitalInput`), and `portfolio_equity_target` was computed but never passed to the optimizer as a constraint at all. Both fields are now threaded through `adapters.py` and used as shown above.

Occupation	HC type	β	EqTarget	RiskyWt	SafeWt	Vol	Sharpe
Biology Professor	bond-like	0.05	+91.6%	91.6%	8.4%	12.1%	0.40
Registered Nurse	bond-like	0.05	+90.6%	90.6%	9.4%	12.0%	0.40
Compliance Officer	bond-like	0.05	+90.8%	90.8%	9.2%	12.0%	0.40
Lawyer	mixed	0.35	+49.9%	49.9%	50.1%	7.8%	0.40
Mechanical Engineer	mixed	0.35	+48.0%	48.0%	52.0%	7.4%	0.40
Financial Analyst	mixed	0.35	+47.8%	47.8%	52.2%	7.4%	0.40
Software Developer	equity-like	0.90	−25.1%	0.0%	100.0%	0.0%	0.00
IT Manager	equity-like	0.90	−25.5%	0.0%	100.0%	0.0%	0.00
Sales Manager	equity-like	0.90	−24.2%	0.0%	100.0%	0.0%	0.00

Table 1: Allocation Agent output for all nine BLS-derived personas: human-capital overlay (BMS/Campbell-Viceira split capped by portfolio equity target) and blended portfolio statistics, first pass (`flag_iteration = 0`). RiskyWt = min(BMS/CV w_{fin}, EqTarget), clipped to $[0, 1]$; Sharpe is invariant to the risky/safe split (≈ 0.40 throughout), as expected when blending with a risk-free asset. [source: `agents/allocation/agent.py`]

Forty-one unit tests pass offline against simulated data (17 for the BL/optimizer core, 20 for the BMS/HC formulas, 4 for concentration constraints); none require WRDS or network access.

Future Directions

- The BMS/Campbell-Viceira formula still clips at 1.0 for extreme H/W ratios rather than surfacing the underlying breakdown: the Software Developer persona's $H/W \approx 25.5\times$ drives the uncapped risky weight to exactly 1.0 before the portfolio equity target intervenes. A bounded formulation would be preferable to a hard clip.
- Incorporate ESG constraints and expand the 24-ticker universe, particularly with single-stock RSU hedge positions.
- Add regime-conditional view scaling so Fama-French view confidence adapts to the Research Agent's detected regime.

3.4 Risk Agent

The Risk Agent (`run_risk_agent()`, deterministic core in `agents/shared/core/risk.py`) stress-tests the Allocation Agent's proposed portfolio against the client's human-capital profile and returns an APPROVE, FLAG, or REJECT decision. Only the reasoning trace is LLM-authored.

Risk pipeline. A 1260-trading-day (5-year) daily returns matrix from CRSP feeds historical VaR and CVaR at the 95% and 99% levels (Basel III/IV convention) and maximum drawdown from the cumulative return path. The current market regime is detected by comparing 60-day rolling volatility to the full-sample baseline (ratio $\geq 1.5\times$: ELEVATED, widens the drawdown cap 25%; $\geq 2.0\times$: CRISIS, widens it 50%) so the system does not force procyclical selling at market bottoms. A dollar-volume-weighted liquidity score and HC-adjusted balance sheet metrics (total wealth, HC fraction, effective equity exposure, employer concentration, economic sector exposures) complete the picture, alongside stress scenarios run against the regime-adjusted cap: S&P 500 2008 (−54% benchmark), COVID 2020 (−34%), 60/40 portfolio 2008 (−23.7%), and an idiosyncratic employer shock (50% shock to employer stock, 30% shock to human capital value).

Decision logic. Concentration limits (single-name 10%, sector 20%, economic-sector 25%, employer 15%) and the risk metrics above drive the APPROVE/FLAG/REJECT decision. Position limits use a static single-name cap labeled `marginal_risk_contribution`, a simplification appropriate for a bounded 24-ticker universe rather than a literal marginal-risk-contribution calculation. Sector limits are HC-correlation-adjusted, adjusted_limit = base_limit$\times(1-\rho)$, where ρ is the client's income-to-equity correlation, applied to the client's own employer sector. On FLAG, tightened constraints feed back to the Allocation Agent for up to three iterations; a critical stress result or an uncured drawdown breach also requests a one-step risk-tolerance downgrade.

Running the full nine-persona pipeline end to end this period (cached CRSP/FRED data, no live network calls) surfaced four correctness bugs, all now fixed and verified against 133/133 passing unit tests. The structural REJECT gate had compared generic human-capital fraction (85 to 97% for every persona in the set, since career value dominates total wealth for most working professionals) against the 15% single-employer concentration limit, rejecting every client regardless of portfolio construction; it is now scaled by `RSU_concentration`, the profile's actual employer-stock signal, so a salaried client with no equity compensation contributes near-zero idiosyncratic employer risk from human capital alone. VaR, CVaR, drawdown, and regime detection had renormalized risky-sleeve weights back to 100% before computing any risk metric, silently discarding the risky/safe split, so a 92%-risky and a 48%-risky client scored identically and a 0%-risky client still showed nonzero drawdown; both are now scaled correctly. A drawdown breach driven by broad market exposure rather than concentration had no working remediation lever, since the FLAG loop only tightened single-name limits, so it now also requests a risk-tolerance downgrade. That downgrade constraint, in turn, was computed by the Risk Agent but never consumed by the Allocation Agent, so a downgrade tightened the Risk Agent's own cap without ever lowering the risky weight it was meant to constrain; `run_allocation()` now reads the same constraint.

Occupation	**RiskyWt**	**VaR95**	**MaxDD**	**Cap**	**Regime**	**Decision**
Biology Professor	91.6%	1.6%	32.9%	15.0%	normal	REJECT (1)
Registered Nurse	90.6%	1.6%	32.6%	15.0%	normal	REJECT (1)
Compliance Officer	90.8%	1.6%	32.6%	15.0%	normal	REJECT (1)
Lawyer	49.9%	0.9%	18.6%	20.0%	normal	REJECT (2)
Mechanical Engineer	48.0%	0.8%	17.5%	20.0%	normal	REJECT (2)
Financial Analyst	47.8%	0.9%	17.9%	15.0%	normal	REJECT (1)
Software Developer	0.0%	−0.0%	−0.0%	25.0%	normal	REJECT (0)
IT Manager	0.0%	−0.0%	−0.0%	25.0%	normal	REJECT (0)
Sales Manager	0.0%	−0.0%	−0.0%	25.0%	normal	REJECT (2)

Table 2: Risk Agent output for all nine BLS-derived personas: VaR/CVaR, drawdown versus the regime-adjusted cap, and the decision after up to three FLAG iterations (parenthetical). Cap is the risk-tolerance drawdown cap (regime-adjusted). All nine converge to REJECT: for bond-like and mixed personas the HC-justified equity share exceeds the drawdown cap tied to stated risk tolerance (see Future Directions); equity-like personas are rejected on the structural employer-concentration gate instead. [source: `agents/risk/agent.py`]

Table 2 exposes a modeling tension rather than a code defect: the HC framework's own math recommends a 91.6% equity share for the bond-like Biology Professor persona, but a portfolio that concentrated has a historical (2008/2020-inclusive) max drawdown near 33%, well above the 15 to 20% cap tied to that client's stated risk tolerance. This is the top priority for weeks 7 through 12, not something patched over under deadline pressure.

Future Directions

- Reconcile the HC-driven equity-target framework with risk-tolerance-based drawdown caps, the central open question raised by this period's end-to-end runs (Table 2).
- Replace the static single-name limit with a true marginal-risk-contribution calculation.
- Add Monte Carlo tail-risk simulation beyond the three historical regime windows, factor-model-based VaR, and dynamic sector limits that respond to the Research Agent's detected regime shifts.
- Add a dedicated Risk Agent unit test suite; unlike the Allocation Agent's 41 tests, no `test_risk_core.py` exists yet.

3.5 Compliance Agent

The Compliance Agent is the final gate before the `AdvisorPackage` is assembled. It is fully deterministic with no LLM calls made, and identical inputs always produce identical outputs, a requirement for regulatory reproducibility. It receives a `ComplianceInput` (assembled by the Orchestrator from three upstream outputs) and a `RiskAgentOutput`, runs eight checks across two structurally distinct jobs, then compiles results into a `ComplianceAgentOutput` via `compile_report()`.

Job 1: Constraint Set Verification (`constraint_checks.py`) audits the Risk Agent's output without recomputing any values. Three functions cover three audit categories:

- **Completeness**: All five academic regime windows present in `regime_evaluation`; every portfolio ticker has a `PositionLimit` entry; all six required `RiskDerivation` fields populated.
- **Consistency**: Cross-checks the Risk Agent's own pass/fail flags against its computed values: `actual_weight` vs. `derived_limit` (positions), `actual_sector_weight` vs. `adjusted_limit` (sectors), `portfolio_drawdown` vs. `drawdown_floor` (regimes). A flag that disagrees with its value is a logical error, elevated to HIGH severity.
- **Derivation Audit**: Verifies correct methodology for this client: equity-like HC ($\beta > 0.8$) must use `hc_correlation_adjusted` sector limits; RSU concentration $> 10\%$ must produce a tighter employer sector cap; `hc_type` in the derivation block must match the Profile Agent's classification.

Job 2: Content & Fiduciary Checks (`content_checks.py`) is Compliance's own independent verification layer. Five functions assess whether the recommendation is suitable, complete, and documented to a fiduciary standard:

- **Rationale Completeness** Every ticker must have a non-empty rationale; absence violates SEC Reg BI's care obligation (17 CFR 240.15l-1). Severity: HIGH.
- **Rationale Specificity**: Each rationale must reference ≥ 2 client-specific terms from a controlled vocabulary (HC type, beta, sector, RSU, regime, risk tolerance). Generic boilerplate fails FINRA Rule 2111. Severity: MEDIUM.
- **HC Acknowledgment**: RSU concentration > 10% requires employer/RSU language (HIGH); $\beta > 0.8$ requires implicit equity exposure language (MEDIUM).
- **Weight Integrity**: Independent re-verification that weights sum to 1.0 ± 0.01, no negatives, at least two positions. Severity: HIGH.
- **Volatility Suitability**: Annualised portfolio volatility must fall within the tolerance band for stated risk tolerance: conservative $[0\%, 12\%]$, moderate $[0\%, 20\%]$, aggressive $[0\%, 30\%]$. Skipped with a pipeline warning if the Risk Agent did not supply `portfolio_volatility_annual`. Severity: HIGH (above band), MEDIUM (below band).

Report compilation (`report.py`) groups violations by `responsible_agent`, determines `compliance_status` (PASS / PASS_WITH_WARNINGS / FAIL), sets `clearance`, and generates a plain-English `recommendation` with specific action items for each agent. A FAIL triggers the Orchestrator's revision loop.

Future Directions

Job 3 (`job3_robo_adviser_checks.py`) will add four checks grounded in SEC IM Guidance Update No. 2017-02, the primary regulatory document governing algorithmic investment advisers under the Investment Advisers Act of 1940:

- Portfolio composition consistent with `investment_objective`
- Algorithm limitation disclosure when `portfolio_equity_target` < 0
- ETF provider conflict-of-interest screen

3.6 Orchestrator Agent

The Orchestrator sequences all five agents and manages two nested feedback loops.

- `run_alloc_risk_loop()`: Runs Allocation then Risk iteratively; FLAG feeds `constraints_violated` back as `flag_constraints`; terminates on APPROVE, REJECT, or after three iterations (third FLAG is contractually invalid and forces REJECT)
- `assemble_compliance_input()`: Denormalizes `ProfileAgentOutput`, `MacroRegimeSnapshot`, and `AllocationAgentOutput` into the flat `ComplianceInput` structure; discretizes `regime_volatility` float into low/medium/high label
- `run_pipeline()`: Top-level entry point; fetches live FRED DGS10 rate (4.4% fallback); runs both feedback loops (Risk: max 3 revisions, Compliance: max 2); bundles all outputs and revision counts into `AdvisorPackage`
- `_revise_allocation_for_compliance()`: Re-runs Allocation with compliance violations as context; increments flag iteration so the agent is aware it is responding to a compliance failure

4 Preliminary Results

As of the mid-term checkpoint, the following components are verified:

- **52 compliance unit tests** pass across all eight checks in Jobs 1 and 2, including edge cases for equity-like personas with RSU concentration, regime stress failures, and malformed Risk Agent outputs.
- **11 profile unit tests** pass, covering the HC annuity formula, effective risk budget, bonus scaling, beta table lookup, and all four Pydantic validators (`total_wealth`, `implicit_equity_exposure`, holdings sum, and HC type / beta consistency).
- The `contracts.py` schema layer correctly validates all `ProfileAgentOutput` instances for all nine BLS personas, enforcing quantitative invariants at construction time.
- The Orchestrator's dual feedback loop logic is implemented and unit-tested against mock agents; FLAG loop termination and compliance revision limits function as specified.
- **41 allocation/HC unit tests** pass, and the full nine-persona Allocation and Risk Agent output (Tables 1 and 2) was generated end to end on cached CRSP/FRED/Fama-French data; 133 unit tests pass across the full pipeline.

```
9 AdvisorPackages produced

        Occupation     HC Type Eq. Target Implicit Exp. Risky % Top Holding   Vol   Risk  FLAG revs         Compliance  Violations  Comp. revs
  Biology Professor   bond-like     +91.6%          4.2%   98.0%  SPY (7.0%) 12.3% REJECT          2               FAIL           1           0
   Registered Nurse   bond-like     +90.6%          4.7%   98.3%  AGG (7.1%) 12.0% REJECT          1               FAIL           1           0
 Compliance Officer   bond-like     +90.8%          4.6%   98.1%  SPY (7.0%) 13.1% REJECT          2               FAIL           1           0
             Lawyer       mixed     +49.9%         31.9%   96.9%  AGG (8.5%) 11.0% REJECT          2               PASS           0           0
Mechanical Engineer       mixed     +48.0%         33.1%   96.9%  AGG (8.4%) 10.9% REJECT          2               PASS           0           0
  Financial Analyst       mixed     +47.8%         33.2%   97.7%  AGG (9.9%) 10.8% REJECT          1               PASS           0           0
 Software Developer equity-like     -25.1%         86.6%  100.0% XLK (10.0%) 15.4% REJECT          2 PASS_WITH_WARNINGS           1           0
         IT Manager equity-like     -25.5%         86.9%  100.0% XLK (10.0%) 15.4% REJECT          2 PASS_WITH_WARNINGS           1           0
      Sales Manager equity-like     -24.2%         86.0%  100.0% SPY (10.0%) 15.3% REJECT          2               PASS           0           0

── Pipeline Warnings ──────────────────────────────────────
  [Biology Professor] Risk Agent REJECTED portfolio after 2 FLAG revision(s)
  [Biology Professor] HIGH violations: check_1_2b_sector_limit_consistency
  [Registered Nurse] Risk Agent REJECTED portfolio after 1 FLAG revision(s)
  [Registered Nurse] HIGH violations: check_1_2b_sector_limit_consistency
  [Compliance Officer] Risk Agent REJECTED portfolio after 2 FLAG revision(s)
  [Compliance Officer] HIGH violations: check_1_2b_sector_limit_consistency
  [Lawyer] Risk Agent REJECTED portfolio after 2 FLAG revision(s)
  [Mechanical Engineer] Risk Agent REJECTED portfolio after 2 FLAG revision(s)
  [Financial Analyst] Risk Agent REJECTED portfolio after 1 FLAG revision(s)
  [Software Developer] Risk Agent REJECTED portfolio after 2 FLAG revision(s)
  [IT Manager] Risk Agent REJECTED portfolio after 2 FLAG revision(s)
  [Sales Manager] Risk Agent REJECTED portfolio after 2 FLAG revision(s)
```

Figure 3: End-to-end pipeline summary across all nine personas.

5 Limitations & Plan for Weeks 7–12

Known limitations:

- Compliance Agent: Job 3 is designed but not yet implemented.
- Allocation Agent: The BMS/Campbell-Viceira risky-weight formula clips at 1.0 for extreme H/W ratios (the Software Developer persona reaches $H/W \approx 25.5\times$) instead of surfacing the breakdown; the 24-ticker ETF universe has no single-stock RSU hedging positions.
- Risk Agent: The HC-driven portfolio equity target and the risk-tolerance-based drawdown cap are not yet reconciled, so HC-theory-justified high-equity allocations for bond-like and mixed personas still breach drawdown caps tied to their stated risk tolerance (Table 2); no dedicated Risk Agent unit test suite exists yet.

Plan for Weeks 7–12:

1. Implement Compliance Agent Job 3 (four SEC robo-adviser checks) and corresponding unit tests.
2. Allocation Agent: expand the ETF universe, add regime-conditional view scaling, and address the BMS clip-at-1.0 breakdown for extreme H/W ratios. Risk Agent: reconcile the HC-driven equity-target framework with risk-tolerance-based drawdown caps, replace the static single-name limit with true marginal-risk-contribution sizing, and add a dedicated unit test suite.
3. Write final paper sections: Introduction, HC Framework, Agent Architecture, Compliance & Regulatory Framework, Results, Conclusion.